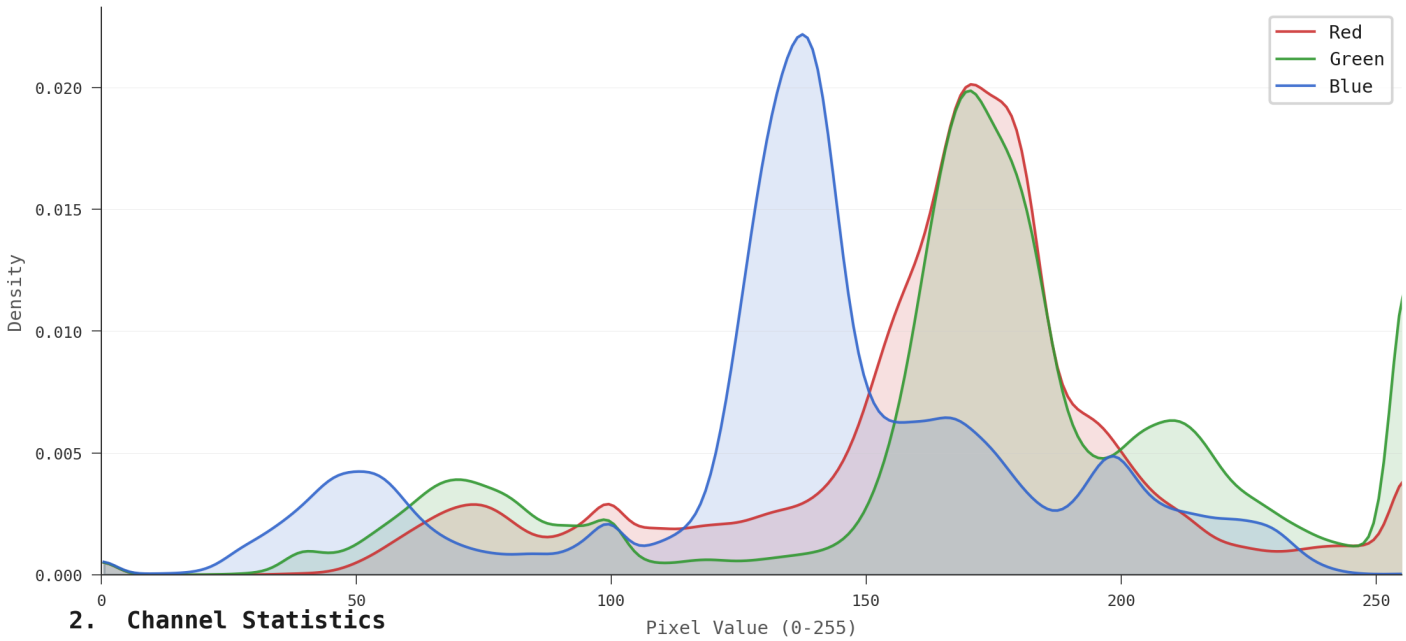
Reference Data Sheet: KODIM12

Subject: Beach Couple Walking

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	161.5734	40.3806	1630.5965	1.1058	-0.7565	0	255
Green	166.9964	50.1228	2512.2992	0.3809	-0.7898	0	255
Blue	137.8578	46.1700	2131.6692	0.2495	-0.4877	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9133	0.8721
G	0.9133	1.0000	0.9674
B	0.8721	0.9674	1.0000

Pairwise Pearson Correlation Coefficients

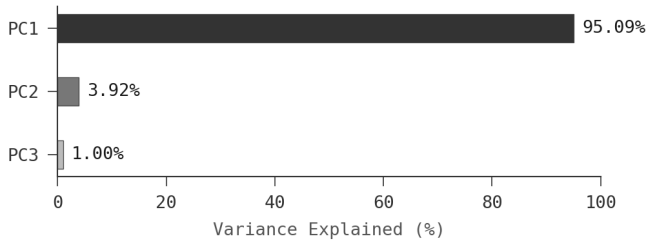
r(R-G)	=	**0.913293**
r(R-B)	=	**0.872110**
r(G-B)	=	**0.967415**
Avg |r|	=	**0.917606**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1630.6007} & 1848.5032 & 1625.9443 \\
1848.5032 & \mathbf{2512.3056} & 2238.7707 \\
1625.9443 & 2238.7707 & \mathbf{2131.6746}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM12

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.4937	**-0.6434**	-0.5850	5966.2169
PC2	-0.8389	0.1752	0.5153	245.8155
PC3	0.2290	-0.7452	0.6263	62.5484

6. Derived Metrics

Condition Number

95.39

(l1/l3) PC1/PC2: 24.27 | PC2/PC3: 3.93

Blue Channel Independence

4.2%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6434 | Green-Blue coupled

Classification: Strongly One-Dimensional (PC1 = 95.09%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.975190	0.959391	0.967290
Green	0.981558	0.971839	0.976698
Blue	0.979094	0.969349	0.974221

8. Average Local Variance (3x3)

Channel	Value
Red	88.4865
Green	101.6072
Blue	91.8665

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 95.39 | PC1: 95.09%

Correlation: G–B pair dominates at r = 0.967415 | R–G: 0.913293 | R–B: 0.872110

Blue Channel: 4.2% independent | 95.8% captured by PC1

Spatial: Avg autocorrelation: 0.972736 | Avg local variance: 93.9867

Eigenvector: Green-Blue coupled | G (-0.6434), B (-0.5850) co-load | R offset (-0.4937)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."